

JECRC University, Jaipur
B.Tech.-I Sem I Year 2025-26
Tutorial Sheet-9 (CO3)
Subject: Mathematics-1
Code: DMA024A

CO3: The tool of power series and Fourier series for learning advanced Engineering Mathematics.

Part-A (2x5=10)

A1. What is an alternating series?

A2. Define Leibnitz's rule for convergence of an alternating series.

A3. What is power series?

A4. Show that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is convergent but not absolutely convergent.

A5. What can you say about the series $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$?

Part –B

(3x7=21)

B1. Find the value of x for which the series $x - \frac{x^2}{2^2} + \frac{x^3}{3^2} - \frac{x^4}{4^2} + \dots$ converges.

B2. Show that an exponential series is convergent for all values of x.

B3. Show $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ is convergent for $-1 \leq x \leq 1$

Part –C

(3x11=33)

C1. Show that Binomial series is convergent for $|x| < 1$.

C2. Find convergence and/or divergence of a power series.

C3. Discuss the uniform convergence of the series $1 + x + x^2 + \dots$

Answers:

A1. A2.	A3.	A4.	A5. absolutely convergent.	B1. [-1,1]
B2.	B3.	C1.	C2.	C3. Uniform convergent in (-1/2, 1/2)